

CSE370 : Database Systems

Assignment 01/02/03 | Spring 2026

ID : <8 Digit Student ID> | Name : Full Official Name

No 1 Query (as Plain Text)	<pre>CREATE TABLE employees (e_id CHAR(4), first_name VARCHAR(20), last_name VARCHAR(20), hire_date DATE, job_id CHAR(10), salary INT, commission_pct DECIMAL(5,3), manager_id CHAR(4), department_id CHAR(3));</pre>
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No 1 SS (of Query & Output in Shell)	<pre> MariaDB [(none)]> create database Company_24301079; Query OK, 1 row affected (0.001 sec) MariaDB [(none)]> use Company_24301079; Database changed MariaDB [Company_24301079]> CREATE TABLE employees (-> e_id CHAR(4), -> first_name VARCHAR(20), -> last_name VARCHAR(20), -> hire_date DATE, -> job_id CHAR(10), -> salary INT, -> commission_pct DECIMAL(5,3), -> manager_id CHAR(4), -> department_id CHAR(3) ->); Query OK, 0 rows affected (0.009 sec) </pre>
No 2 Query (as Plain Text)	<pre> insert into employees value ('E001','John','Doe','2020-01-15','IT_PROG',65000,NULL,'E010','D01'), ('E002','Jane','Smith','2019-03-22','SA_REP',55000,0.150,'E011','D02'), ('E003','Michael','Johnson','2021-06-10','IT_PROG',62000,NULL,'E010','D01'), ('E004','Emily','Davis','2018-11-05','SA_REP',58000,0.200,'E011','D02'), ('E005','David','Wilson','2022-02-14','IT_PROG',60000,NULL,'E010','D01'), ('E006','Sarah','Brown','2020-07-30','HR_REP',50000,0.150,'E012','D03'), ('E007','James','Taylor','2019-09-18','SA_REP',60000,0.180,'E011','D02'), ('E008','Lisa','Anderson','2021-04-25','FI_ACC',52000,NULL,'E013','D04'), ('E009','Robert','Martinez','2020-12-01','IT_PROG',67000,NULL,'E010','D01'), ('E010','Patricia','Garcia','2017-05-10','IT_MAN',85000,NULL,NULL,'D01'), </pre>

	<pre>('E011', 'Thomas', 'Lee', '2016-08-20', 'SA_MAN', 80000, 0.100, NULL, 'D02'), ('E012', 'Maria', 'White', '2018-02-15', 'HR_MAN', 75000, NULL, NULL, 'D03'), ('E013', 'Daniel', 'Harris', '2019-10-30', 'FI_MAN', 78000, NULL, NULL, 'D04'), ('E014', 'Jessica', 'Clark', '2021-01-20', 'HR_REP', 48000, 0.100, 'E012', 'D03'), ('E015', 'Kevin', 'Lewis', '2022-03-12', 'FI_ACC', 51000, NULL, 'E013', 'D04');</pre>
No 2 SS (of Query & Output in Shell)	<pre>MariaDB [Company_24301079]> insert into employees value -> ('E001', 'John', 'Doe', '2020-01-15', 'IT_PROG', 65000, NULL, 'E010', 'D01'), -> ('E002', 'Jane', 'Smith', '2019-03-22', 'SA_REP', 55000, 0.150, 'E011', 'D02'), -> ('E003', 'Michael', 'Johnson', '2021-06-10', 'IT_PROG', 62000, NULL, 'E010', 'D01'), -> ('E004', 'Emily', 'Davis', '2018-11-05', 'SA_REP', 58000, 0.200, 'E011', 'D02'), -> ('E005', 'David', 'Wilson', '2022-02-14', 'IT_PROG', 60000, NULL, 'E010', 'D01'), -> ('E006', 'Sarah', 'Brown', '2020-07-30', 'HR_REP', 50000, 0.150, 'E012', 'D03'), -> ('E007', 'James', 'Taylor', '2019-09-18', 'SA_REP', 60000, 0.180, 'E011', 'D02'), -> ('E008', 'Lisa', 'Anderson', '2021-04-25', 'FI_ACC', 52000, NULL, 'E013', 'D04'), -> ('E009', 'Robert', 'Martinez', '2020-12-01', 'IT_PROG', 67000, NULL, 'E010', 'D01'), -> ('E010', 'Patricia', 'Garcia', '2017-05-10', 'IT_MAN', 85000, NULL, NULL, 'D01'), -> ('E011', 'Thomas', 'Lee', '2016-08-20', 'SA_MAN', 80000, 0.100, NULL, 'D02'), -> ('E012', 'Maria', 'White', '2018-02-15', 'HR_MAN', 75000, NULL, NULL, 'D03'), -> ('E013', 'Daniel', 'Harris', '2019-10-30', 'FI_MAN', 78000, NULL, NULL, 'D04'), -> ('E014', 'Jessica', 'Clark', '2021-01-20', 'HR_REP', 48000, 0.100, 'E012', 'D03'), -> ('E015', 'Kevin', 'Lewis', '2022-03-12', 'FI_ACC', 51000, NULL, 'E013', 'D04'); Query OK, 15 rows affected (0.007 sec) Records: 15 Duplicates: 0 Warnings: 0</pre>
No 3 Query (as Plain Text)	<pre>SELECT first_name, last_name, hire_date, department_id FROM employees WHERE hire_date = (SELECT MAX(hire_date) FROM employees);</pre>

No 3 SS (of Query & Output in Shell)	<pre> MariaDB [Company_24301079]> SELECT first_name, last_name, hire_date, department_id -> FROM employees -> WHERE hire_date = (SELECT MAX(hire_date) FROM employees); +-----+-----+-----+-----+ first_name last_name hire_date department_id +-----+-----+-----+-----+ Kevin Lewis 2022-03-12 D04 +-----+-----+-----+-----+ 1 row in set (0.000 sec) </pre>
No 4 Query (as Plain Text)	<pre> SELECT first_name, last_name, e_id, salary, department_id FROM employees e WHERE salary = (SELECT MAX(salary) FROM employees WHERE department_id = e.department_id); </pre>

No 4 SS
(of Query & Output
in Shell)

```
MariaDB [Company_24301079]> SELECT first_name, last_name, e_id, salary, department_id  
-> FROM employees e  
-> WHERE salary = (  
->     SELECT MAX(salary)  
->     FROM employees  
->     WHERE department_id = e.department_id  
-> );
```

first_name	last_name	e_id	salary	department_id
Patricia	Garcia	E010	85000	D01
Thomas	Lee	E011	80000	D02
Maria	White	E012	75000	D03
Daniel	Harris	E013	78000	D04

4 rows in set (0.000 sec)

No 5 Query
(as Plain Text)

```
SELECT first_name, last_name, e_id, commission_pct, dept_id  
FROM employees  
WHERE dept_id = 'D02'  
AND commission_pct > ALL (  
    SELECT commission_pct  
    FROM employees  
    WHERE dept_id = 'D01'  
    AND commission_pct IS NOT NULL  
);
```

No 5 SS
(of Query & Output
in Shell)

```
ERROR 1054 (42S22): unknown column 'dept_id' in 'field list'
MariaDB [Company_24301079]> SELECT first_name, last_name, e_id, commission_pct
, department_id
  -> FROM employees
  -> WHERE department_id = 'D02'
  -> AND commission_pct > ALL (
  ->   SELECT commission_pct
  ->   FROM employees
  ->   WHERE department_id = 'D01'
  ->   AND commission_pct IS NOT NULL
  -> );
```

first_name	last_name	e_id	commission_pct	department_id
Jane	Smith	E002	0.150	D02
Emily	Davis	E004	0.200	D02
James	Taylor	E007	0.180	D02
Thomas	Lee	E011	0.100	D02

4 rows in set (0.001 sec)

No 6 Query
(as Plain Text)

```
SELECT department_id, COUNT(*) AS total_employees
FROM employees
GROUP BY department_id
HAVING MAX(salary) <= 70000;
```

No 6 SS (of Query & Output in Shell)	<pre> MariaDB [Company_24301079]> SELECT department_id, COUNT(*) AS total_employees -> FROM employees -> GROUP BY department_id -> HAVING MAX(salary) <= 70000; Empty set (0.000 sec) </pre>
No 7 Query (as Plain Text)	<pre> SELECT e1.department_id, e1.job_id, e1.salary FROM employees e1 WHERE e1.salary < ANY (SELECT e2.salary FROM employees e2 WHERE e2.department_id = e1.department_id AND e2.job_id <> e1.job_id); </pre>

No 7 SS
(of Query & Output
in Shell)

```
MariaDB [Company_24301079]> SELECT e1.department_id, e1.job_id, e1.salary  
-> FROM employees e1  
-> WHERE e1.salary < ANY (  
->     SELECT e2.salary  
->     FROM employees e2  
->     WHERE e2.department_id = e1.department_id  
->     AND e2.job_id <> e1.job_id  
-> );
```

department_id	job_id	salary
D01	IT_PROG	65000
D02	SA_REP	55000
D01	IT_PROG	62000
D02	SA_REP	58000
D01	IT_PROG	60000
D03	HR_REP	50000
D02	SA_REP	60000
D04	FI_ACC	52000
D01	IT_PROG	67000
D03	HR_REP	48000
D04	FI_ACC	51000

11 rows in set (0.001 sec)

No 8 Query
(as Plain Text)

```
SELECT manager_id  
FROM employees  
GROUP BY manager_id  
HAVING MIN(salary) >= 55000;
```


No 8 SS (of Query & Output in Shell)	<pre> MariaDB [Company_24301079]> SELECT manager_id -> FROM employees -> GROUP BY manager_id -> HAVING MIN(salary) >= 55000; +-----+ manager_id +-----+ NULL E010 E011 +-----+ 3 rows in set (0.000 sec) </pre>
No 9 Query (as Plain Text)	<pre> SELECT first_name, last_name, e_id, salary, department_id, commission_pct FROM employees e1 WHERE commission_pct = (SELECT MAX(e2.commission_pct) FROM employees e2 WHERE e2.manager_id = e1.manager_id); </pre>

No 9 SS
(of Query & Output
in Shell)

```
MariaDB [Company_24301079]> SELECT first_name, last_name, e_id, salary, depart  
ment_id, commission_pct  
-> FROM employees e1  
-> WHERE commission_pct = (  
->     SELECT MAX(e2.commission_pct)  
->     FROM employees e2  
->     WHERE e2.manager_id = e1.manager_id  
-> );
```

first_name	last_name	e_id	salary	department_id	commission_pct
Emily	Davis	E004	58000	D02	0.200
Sarah	Brown	E006	50000	D03	0.150

```
2 rows in set (0.001 sec)
```

No 10 Query
(as Plain Text)

```
SELECT job_id, AVG(salary) AS avg_salary  
FROM employees  
GROUP BY job_id  
HAVING AVG(salary) < (  
    SELECT AVG(salary)  
    FROM employees  
);
```

No 10 SS
(of Query & Output
in Shell)

```
MariaDB [Company_24301079]> SELECT job_id, AVG(salary) AS avg_salary  
-> FROM employees  
-> GROUP BY job_id  
-> HAVING AVG(salary) < (  
->     SELECT AVG(salary)  
->     FROM employees  
-> );
```

```
+-----+-----+  
| job_id | avg_salary |  
+-----+-----+  
| FI_ACC | 51500.0000 |  
| HR_REP | 49000.0000 |  
| SA_REP | 57666.6667 |  
+-----+-----+  
3 rows in set (0.001 sec)
```